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Tetrapad I2C Specs

Eurorack
tetrapad

kamil #1 March 1, 2021, 9:30pm

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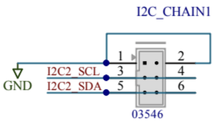
Some people have expressed interest in integrating the Tetrapad with other I2C devices such as those made by Monome or Orthogonal Devices. Unfortunately we're too busy with other projects here to be able to do such integrations ourselves so with that in mind I'm going to publish some info here that can help others do that. I'll be monitoring this thread to answer any questions / provide additional information as needed.

Hardware Info

The I2C connectors on the Tetrapad are both the same, there are two to facilitate easy chaining between devices. The pinout is compatible with Monome's ii bus:



The address of the Tetrapad is selected via the switches on the back of the module, allowing up to four Tetrapads to be used on one I2C bus. With both switches in the off (00) position the address is 0x3B. Using the switches it's possible to set the address to 0x3C (01), 0x3D (10), or 0x3E (11).

The Tetrapad I2C interface works on a pretty typical request-response basis like most I2C devices.
Each command consists of an initial byte that identifies the specific command, then a variable number of bytes of data, depending on the command.

Commands

Here's the available commands, posted as C++ code. I may massage these into a table later but hopefully this is clear enough.

```
// Tetrapad outputs and LEDs will be updated only via I2C requests.
constexpr auto ENABLE_HANDLER = Info{ID::ENABLE_HANDLER, 0, 1};

// Disable the I2C handler on the Tetrapad, re-enabling all native functionali
constexpr auto DISABLE_HANDLER = Info{ID::DISABLE_HANDLER, 0, 1};

// Read the sensor values.
// The values are returned as two packed arrays of 12-bit numbers plus an 8-bi
// The first array is the last-touched position of each sensor.
// The second array is the current pressure of each sensor.
// The bitmask indicates which sensors are currently being touched.
constexpr auto READ_SENSORS = Info{ID::READ_SENSORS, 0, IJN::Packed12BitSize(C

// Read the button and encoder states.
// The first byte contains a bitmask for the button states with the following
// 0 - 3: Encoders
// 4: Shift (red button)
// 5: Config (white button)
// The next four bytes are the current position of the encoder, from 0-255.
constexpr auto READ_UI = Info{ID::READ_UI, 0, 5};

// Set the output values.
// Each output is 12-bit, with 0xfff / 2 being 0V.
constexpr auto SET_OUTPUTS = Info{ID::SET_OUTPUTS, IJN::Packed12BitSize(Consta
```

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stretta #2 March 7, 2021, 4:28pm

This is great information. Thank you for providing it. Would the I2C on the 1U MIDI also be worth getting into?

kamil #3 March 8, 2021, 12:27am

The MIDI 1U is an I2C host, not a device, so it can't respond to I2C messages.

The CVx expander for it is an I2C device (actually two I2C devices). I could provide the specs for those if you are interested.

1 Like

stretta #4 March 8, 2021, 12:52am

Sorry. Yes, the CVx. It's really elegant.

kamil #5 March 8, 2021, 12:57am

Ok I'll make a post about that as well. I feel like I already wrote it up somewhere, but that may have been an email to someone.

Edit: ok, I did actually post about this in the CVx thread:

> CVx - CV Expander compatible with the MIDI 1U Eurorack
>
> Sure, here's the info. Each CVX consists of two two I2C devices, one EEPROM and one DAC. The DAC is an LTC2605, so you can find the necessary commands in its datasheet. The EEPROM just uses the standard I2C eeprom command set. Addresses for each based on the CVx address jumper settings are: EEPROM Device Address Exp 0 0x53 Exp 1 0x51 Exp 2 0x52 Exp 3 0x50 DAC (LTC2605) Device Address Exp 0 0x30 Exp 1 0x12 Exp 2 0x22 Exp 3 0x10 The EEPROM contains the calibratio…

Hopefully that's enough info to get you going, let me know if you have any questions!

1 Like

atomboyd #6 May 5, 2021, 9:42pm

Fascinating! Thanks to both of you for all this info. It looks like the 1U row on my 2nd stealth case just took an interesting turn. Great stuff!

1 Like